Phase 3 Plan: Telemetry

A simple explanation of what's next, and how we're going to tackle it — September 2026

What Is Phase 3 About?

Phases 1 and 2 are done: we can read an FIA penalty document and save the details (who was involved, what corner, what the verdict was) neatly into a database. But right now that's just words on a page — a document *says* “Car 77 collided with Car 20 in turn 6,” but we don't have any actual race data proving what really happened at that moment: how fast each car was going, where exactly they were on track, who was really ahead.

That's what Phase 3 is for. **Telemetry** is the real, measured data recorded during a race — speed, braking, throttle, and exact position on track, many times per second, for every car. A tool called **FastF1** gives us free access to this data for past races. The goal of Phase 3 is simple to say and hard to do: **for each penalty we've already parsed, go find the matching moment in the real telemetry data.**

Why Is This the Hard Part?

Here's the actual problem, in plain terms: the FIA document and FastF1 each describe “when something happened” in a completely different language.

- The FIA document says something like **“17:49”** — an actual clock time, like a wall clock on the garage wall.
- FastF1 organizes everything by **lap number** and **time since the session started** (like a stopwatch that started at the green flag) — not by clock time.

So before we can pull “the telemetry at 17:49,” we first have to figure out which lap and which stopwatch-moment that clock time actually corresponds to. This is genuinely an open problem right now — we haven't solved it yet, and it may take some trial and error once we're looking at real data side by side. This plan lays out how we're going to approach it, not a guarantee of exactly how it'll turn out.

Good news: FastF1 does record real clock timestamps internally alongside its lap/stopwatch data (every telemetry sample and every lap has an actual date-and-time attached, not just a lap number) — so the connection we need almost certainly exists in the data. The work is figuring out how to reliably line the two time formats up, and handling cases where they don't match perfectly (e.g. time zones, or FIA rounding to the nearest minute).

The Plan: Steps We'll Take

Here's the order we'll tackle this in. Each step builds on the one before it, and we'll test with one real example first before trying to automate it for every parsed decision.

1. **Pick one already-parsed decision to test with.** We already have four real ones saved in the database from Phase 1 & 2 — we'll use one of those as our guinea pig rather than starting with something new.
2. **Figure out which race/session it belongs to.** The FIA filename and document tell us the year and Grand Prix (e.g. “2024 Abu Dhabi Grand Prix”) and the session (“Race”). We'll turn that into the inputs FastF1 needs.
3. **Load that session with FastF1** and pull up the driver's laps using the car number we already extracted (`own_car`).
4. **Try to line up the FIA clock time with FastF1's timestamps** — this is the real unknown. We'll look at the actual timestamps FastF1 gives us for that driver's laps around the incident, and see how closely we can match “17:49” to one of them.
5. **Once we find the right moment, pull the telemetry** — speed, position, throttle, brake — for both cars involved (`own_car` and `other_car`) around that moment.
6. **Check it against what the document says happened.** Does the telemetry actually show two cars close together, braking hard, near the turn number the document mentions? This sanity check is how we'll know the matching actually worked, before trusting it for other decisions.

■ **What “done” looks like for this phase:** one real penalty decision, with real telemetry pulled up alongside it, that we can look at and say “yes, this is clearly the same moment the document is describing.” We are not trying to handle every decision automatically yet — that comes later, once the approach is proven on one example.

Technical Detail: What We're Starting From

We already have one proof-of-concept script, written before Phase 1 even started: `backend/test_fastf1.py`. It doesn't connect to any penalty decision yet — it just proves FastF1 works and shows the shape of the data we'll be working with:

```
import fastf1

fastf1.Cache.enable_cache("cache")

session = fastf1.get_session(2023, "Bahrain", "R")
session.load()

# Get Verstappen's fastest lap
ver_lap = session.laps.pick_driver("VER").pick_fastest()

# Get telemetry for that lap
telemetry = ver_lap.get_telemetry()

print(telemetry[["Speed", "X", "Y", "Distance", "Throttle", "Brake"]].head(20))
```

fastf1.get_session(year, event, session_type) — loads one race weekend's session (“R” for Race). This maps directly to fields we already have: the year and Grand Prix name come from the FIA document, and `result["Session"]` already stores “Race”.

session.laps.pick_driver(code) — FastF1 identifies drivers by a 3-letter code (like “VER”), not by car number. One small task inside this phase: build a lookup from car number (which is what we have, e.g. `own_car = 77`) to driver code (e.g. “BOT”) — FastF1's session data includes this mapping already.

lap.get_telemetry() — returns a table of measurements (speed, position, throttle, brake, and more) taken many times per second throughout that lap, each with its own real timestamp attached. This timestamp is exactly what we'll try to match against the FIA document's `incident_time` field.

A Couple of Terms, Explained Simply

Telemetry — real measured data recorded during a race (speed, position, braking, and more), many times per second, for every car.

Session — one distinct part of a race weekend (Practice, Qualifying, or the Race itself). FastF1 organizes all its data by session.

Wall-clock time — an actual time of day, like “17:49”, the way FIA writes incident times.

Lap-relative / session-relative time — time measured from a starting point (like a stopwatch started at the green flag), the way FastF1 organizes most of its data by default.

That's the plan. Once we prove this works for one real decision, the rest of Phase 3 is repeating the same steps for the other parsed decisions — and then we're ready for Phase 4, actually attaching this telemetry to each penalty record permanently.